Biochemistry Cume **November 12, 2010**

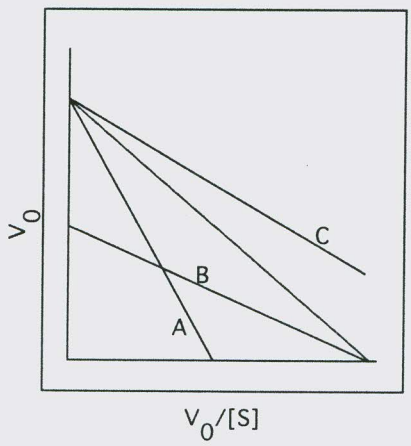
Enzyme Behavior

1. When enzyme solutions are heated, there is a progressive loss of catalytic activity over time due to denaturation of the enzyme. A solution of the enzyme xanthine oxidase was incubated at 50° C and lost about 65% of its activity in 10 minutes. However, when this enzyme solution was heated at the same temperature in the presence of excess of one of its substrates, it lost approximately 5% of its activity over the same time span. Write a short paragraph that will justify the rationale for this experimental observation. (40 pts.)

2. One transformation of the Michaelis-Menten equation is the Lineweaver-Burk (LB) equation. Multiplying both sides of the LB equation by V_{max} and rearranging will give the Eadie-Hofstee equation:

$$V_0 = (-K_m)\left(\frac{V_0}{S}\right) + V_{max}$$

A plot of V_0 vs. $V_0/[S]$ for an enzyme-catalyzed reaction is shown below.



The red curve on the graph above was obtained in the absence of inhibitor. Which of the other curves (A, B, or C) shows the enzyme activity when a competitive inhibitor is added to the reaction mixture? Explain your answer in detail. (30 pts.)

3. Below is the Michaelis-Menten relationship.

$$V = \frac{V_{max}[S]}{K_M + [S]}$$

For an enzyme that displays Michaelis-Menten kinetics, what is the reaction velocity, V, as a percentage of V_{max} observed when $[S] = (0.1)K_M$. (30 pts).